Homework 4:

In homework 4, I first design the full adder in matlab Simulink and use this block to design a ripple carry adder. After designing, I generate VHDL code and install it DE-soc1 board.

Screenshot of MATLAB circuit:

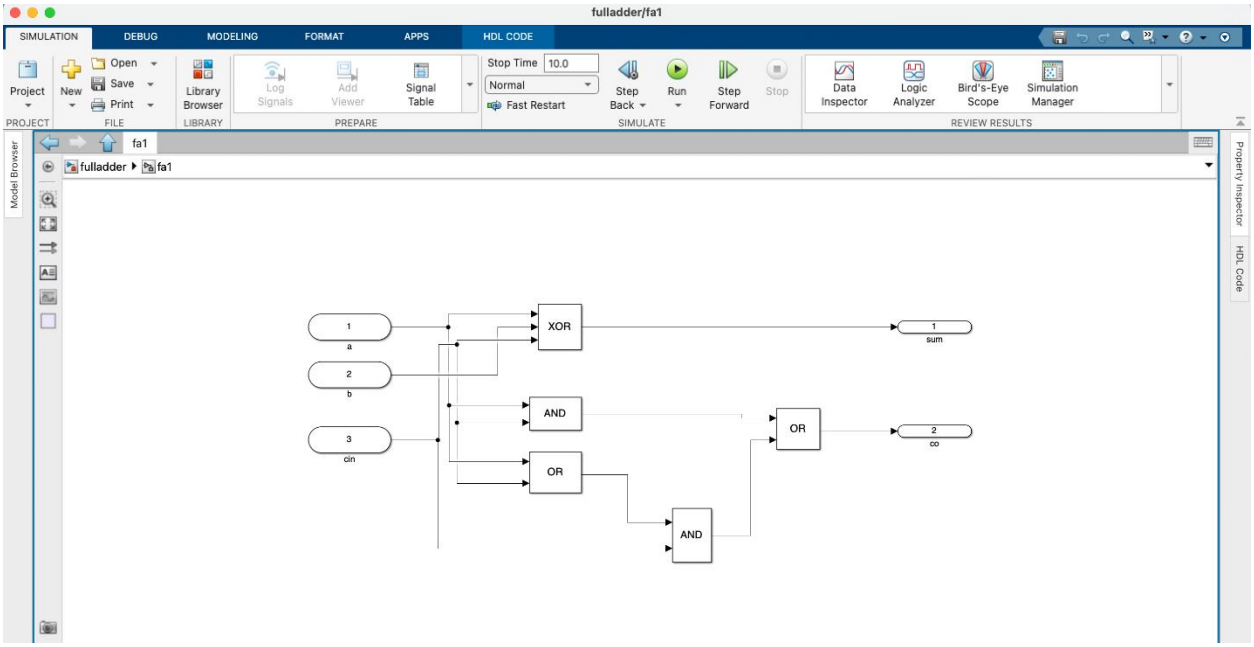


Figure: full adder in MATLAB Simulink

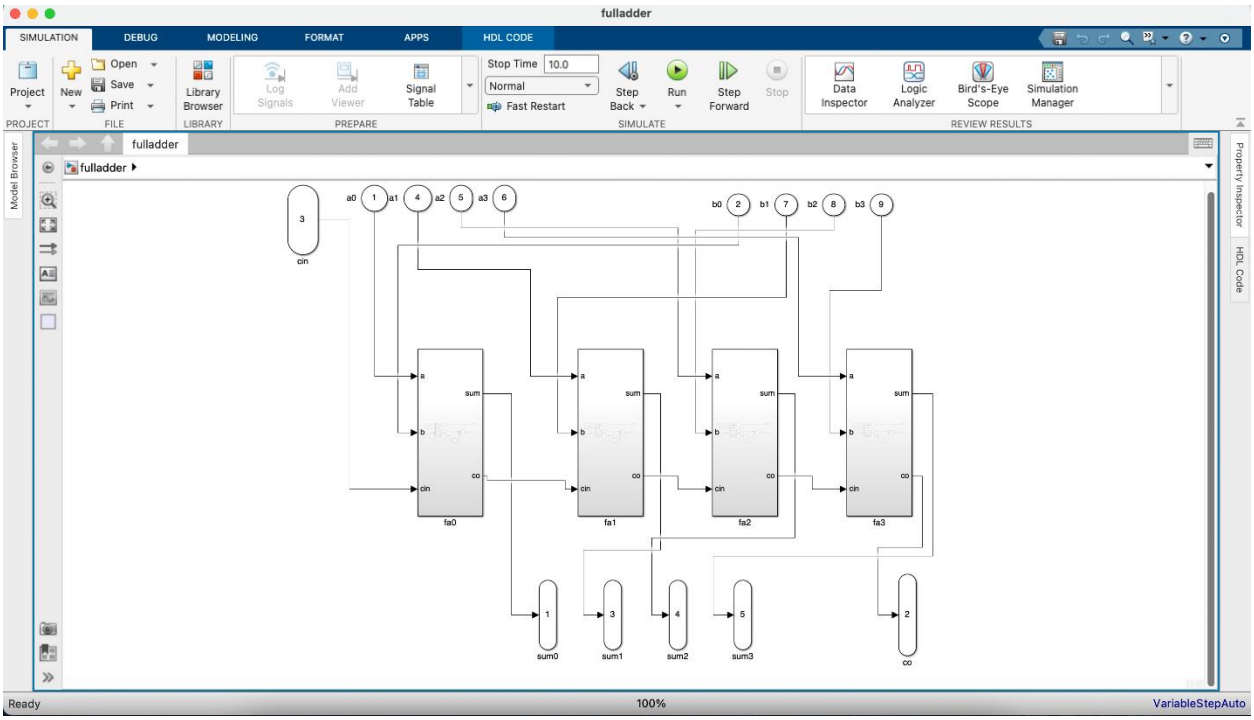
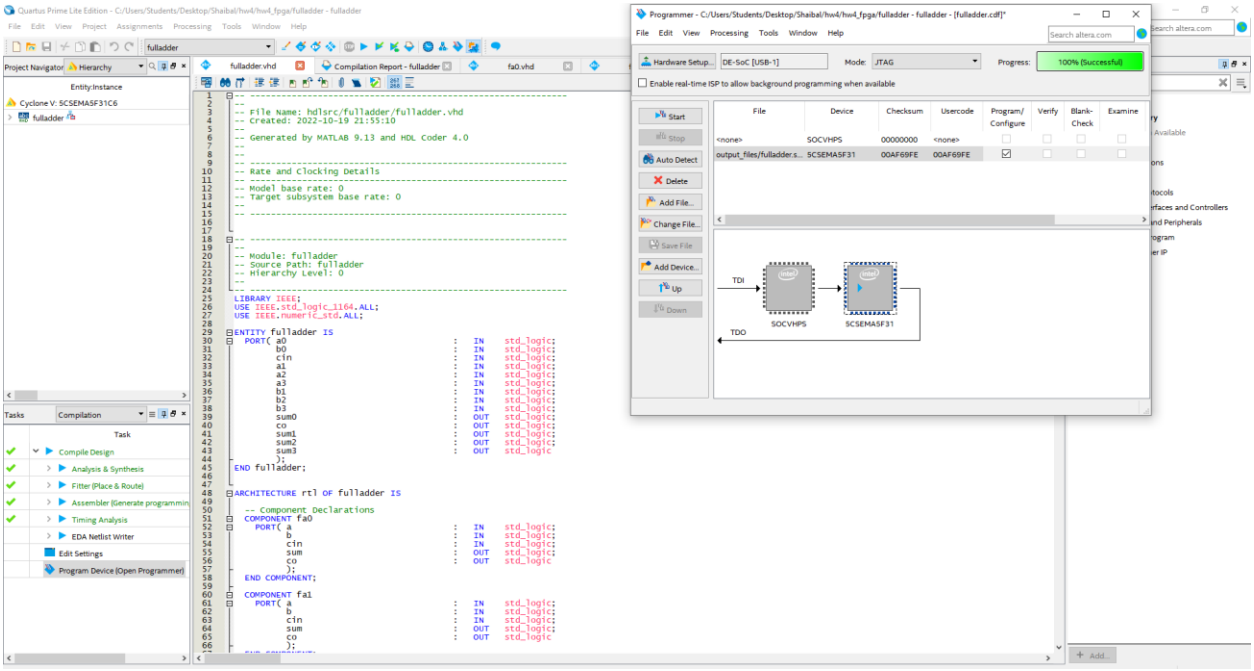


Figure: ripple carry adder in MATLAB Simulink

The generated VHDL codes screenshots are below:



```
  END COMPONENT;

  COMPONENT fa1
    PORT( a                               :   IN    std_logic;
          b                               :   IN    std_logic;
          cin                             :   IN    std_logic;
          sum                             :   OUT   std_logic;
          co                              :   OUT   std_logic
          );
  END COMPONENT;

  COMPONENT fa2
    PORT( a                               :   IN    std_logic;
          b                               :   IN    std_logic;
          cin                             :   IN    std_logic;
          sum                             :   OUT   std_logic;
          co                              :   OUT   std_logic
          );
  END COMPONENT;

  COMPONENT fa3
    PORT( a                               :   IN    std_logic;
          b                               :   IN    std_logic;
          cin                             :   IN    std_logic;
          sum                             :   OUT   std_logic;
          co                              :   OUT   std_logic
          );
  END COMPONENT;

  -- Component Configuration Statements
  FOR ALL : fa0
    USE ENTITY work.fa0(rtl);

  FOR ALL : fa1
    USE ENTITY work.fa1(rtl);

  FOR ALL : fa2
    USE ENTITY work.fa2(rtl);

  FOR ALL : fa3
    USE ENTITY work.fa3(rtl);

  -- Signals
  SIGNAL fa0_out1                         : std_logic;
  SIGNAL fa0_out2                         : std_logic;
  SIGNAL fa1_out1                         : std_logic;
  SIGNAL fa1_out2                         : std_logic;
  SIGNAL fa2_out1                         : std_logic;
  SIGNAL fa2_out2                         : std_logic;
  SIGNAL fa3_out1                         : std_logic;
  SIGNAL fa3_out2                         : std_logic;
```

```
BEGIN
  u_fa0 : fa0
    PORT MAP( a => a0,
              b => b0,
              cin => cin,
              sum => fa0_out1,
              co => fa0_out2
              );

  u_fa1 : fa1
    PORT MAP( a => a1,
              b => b1,
              cin => fa0_out2,
              sum => fa1_out1,
              co => fa1_out2
              );

  u_fa2 : fa2
    PORT MAP( a => a2,
              b => b2,
              cin => fa1_out2,
              sum => fa2_out1,
              co => fa2_out2
              );

  u_fa3 : fa3
    PORT MAP( a => a3,
              b => b3,
              cin => fa2_out2,
              sum => fa3_out1,
              co => fa3_out2
              );

  sum0 <= fa0_out1;

  co <= fa3_out2;

  sum1 <= fa1_out1;

  sum2 <= fa2_out1;

  sum3 <= fa3_out1;

END rtl;
```

Figure: ripple carry adder main function generated from MATLAB Simulink

The following fa0,fa1,fa2,fa3 full adder blocks functions are below:

fulladder.vhd | Compilation Report - fulladder | fa0.vhd | fa1.vhd | fa2.vhd | fa3.vhd

```
-- -------------------------------------------------------------
--
-- File Name: hdlsrc/fulladder/fa0.vhd
-- Created: 2022-10-19 21:55:10
--
-- Generated by MATLAB 9.13 and HDL Coder 4.0
--
-- -------------------------------------------------------------


-- -------------------------------------------------------------
--
-- Module: fa0
-- Source Path: fulladder/fa0
-- Hierarchy Level: 1
--
-- -------------------------------------------------------------
LIBRARY IEEE;
USE IEEE.std_logic_1164.ALL;
USE IEEE.numeric_std.ALL;

ENTITY fa0 IS
  PORT( a                                 :   IN    std_logic;
        b                                 :   IN    std_logic;
        cin                               :   IN    std_logic;
        sum                               :   OUT   std_logic;
        co                                :   OUT   std_logic
        );
END fa0;


ARCHITECTURE rtl OF fa0 IS

  -- Signals
  SIGNAL Logical_Operator1_out1           : std_logic;
  SIGNAL Logical_Operator5_out1           : std_logic;
  SIGNAL Logical_Operator3_out1           : std_logic;
  SIGNAL Logical_Operator2_out1           : std_logic;
  SIGNAL Logical_Operator4_out1           : std_logic;

BEGIN
  Logical_Operator1_out1 <= cin XOR (a XOR b);

  Logical_Operator5_out1 <= a AND cin;

  Logical_Operator3_out1 <= a OR cin;

  Logical_Operator2_out1 <= Logical_Operator3_out1 AND cin;

  Logical_Operator4_out1 <= Logical_Operator5_out1 OR Logical_Operator2_out1;

  sum <= Logical_Operator1_out1;

  co <= Logical_Operator4_out1;

END rtl;
```

fulladder.vhd | Compilation Report - fulladder | fa0.vhd | fa1.vhd | fa2.vhd | fa3.vhd

```
-- -------------------------------------------------------------
--
-- File Name: hdlsrc/fulladder/fa1.vhd
-- Created: 2022-10-19 21:55:10
--
-- Generated by MATLAB 9.13 and HDL Coder 4.0
--
-- -------------------------------------------------------------


-- -------------------------------------------------------------
--
-- Module: fa1
-- Source Path: fulladder/fa1
-- Hierarchy Level: 1
--
-- -------------------------------------------------------------
LIBRARY IEEE;
USE IEEE.std_logic_1164.ALL;
USE IEEE.numeric_std.ALL;

ENTITY fa1 IS
  PORT( a                                 :   IN    std_logic;
        b                                 :   IN    std_logic;
        cin                               :   IN    std_logic;
        sum                               :   OUT   std_logic;
        co                                :   OUT   std_logic
        );
END fa1;


ARCHITECTURE rtl OF fa1 IS

  -- Signals
  SIGNAL Logical_Operator1_out1           : std_logic;
  SIGNAL Logical_Operator5_out1           : std_logic;
  SIGNAL Logical_Operator3_out1           : std_logic;
  SIGNAL Logical_Operator2_out1           : std_logic;
  SIGNAL Logical_Operator4_out1           : std_logic;

BEGIN
  Logical_Operator1_out1 <= cin XOR (a XOR b);

  Logical_Operator5_out1 <= a AND cin;

  Logical_Operator3_out1 <= a OR cin;

  Logical_Operator2_out1 <= Logical_Operator3_out1 AND cin;

  Logical_Operator4_out1 <= Logical_Operator5_out1 OR Logical_Operator2_out1;

  sum <= Logical_Operator1_out1;

  co <= Logical_Operator4_out1;

END rtl;
```

fulladder.vhd | Compilation Report - fulladder | fa0.vhd | fa1.vhd | fa2.vhd | fa3.vhd

```
-- -------------------------------------------------------------
--
-- File Name: hdlsrc/fulladder/fa2.vhd
-- Created: 2022-10-19 21:55:10
--
-- Generated by MATLAB 9.13 and HDL Coder 4.0
--
-- -------------------------------------------------------------


-- -------------------------------------------------------------
--
-- Module: fa2
-- Source Path: fulladder/fa2
-- Hierarchy Level: 1
--
-- -------------------------------------------------------------
LIBRARY IEEE;
USE IEEE.std_logic_1164.ALL;
USE IEEE.numeric_std.ALL;

ENTITY fa2 IS
  PORT( a                                 :   IN    std_logic;
        b                                 :   IN    std_logic;
        cin                               :   IN    std_logic;
        sum                               :   OUT   std_logic;
        co                                :   OUT   std_logic
        );
END fa2;


ARCHITECTURE rtl OF fa2 IS

  -- Signals
  SIGNAL Logical_Operator1_out1           : std_logic;
  SIGNAL Logical_Operator5_out1           : std_logic;
  SIGNAL Logical_Operator3_out1           : std_logic;
  SIGNAL Logical_Operator2_out1           : std_logic;
  SIGNAL Logical_Operator4_out1           : std_logic;

BEGIN
  Logical_Operator1_out1 <= cin XOR (a XOR b);

  Logical_Operator5_out1 <= a AND cin;

  Logical_Operator3_out1 <= a OR cin;

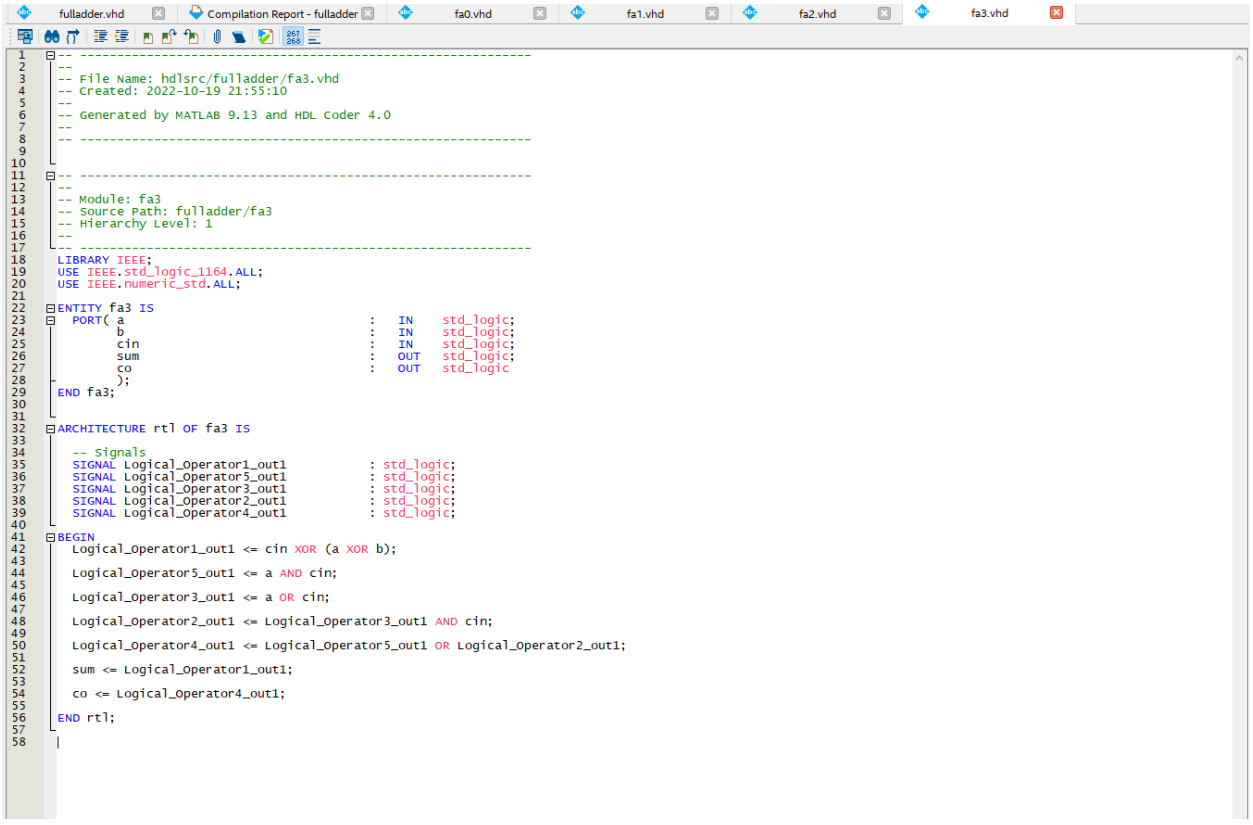
  Logical_Operator2_out1 <= Logical_Operator3_out1 AND cin;

  Logical_Operator4_out1 <= Logical_Operator5_out1 OR Logical_Operator2_out1;

  sum <= Logical_Operator1_out1;

  co <= Logical_Operator4_out1;

END rtl;
```



```
fulladder.vhd | Compilation Report - fulladder | fa0.vhd | fa1.vhd | fa2.vhd | fa3.vhd

-- -------------------------------------------------------------
--
-- File Name: hdlsrc/fulladder/fa3.vhd
-- Created: 2022-10-19 21:55:10
--
-- Generated by MATLAB 9.13 and HDL Coder 4.0
--
-- -------------------------------------------------------------


-- -------------------------------------------------------------
--
-- Module: fa3
-- Source Path: fulladder/fa3
-- Hierarchy Level: 1
--
-- -------------------------------------------------------------
LIBRARY IEEE;
USE IEEE.std_logic_1164.ALL;
USE IEEE.numeric_std.ALL;

ENTITY fa3 IS
  PORT( a                                 :   IN    std_logic;
        b                                 :   IN    std_logic;
        cin                               :   IN    std_logic;
        sum                               :   OUT   std_logic;
        co                                :   OUT   std_logic
        );
END fa3;


ARCHITECTURE rtl OF fa3 IS

  -- Signals
  SIGNAL Logical_Operator1_out1           : std_logic;
  SIGNAL Logical_Operator5_out1           : std_logic;
  SIGNAL Logical_Operator3_out1           : std_logic;
  SIGNAL Logical_Operator2_out1           : std_logic;
  SIGNAL Logical_Operator4_out1           : std_logic;

BEGIN
  Logical_Operator1_out1 <= cin XOR (a XOR b);

  Logical_Operator5_out1 <= a AND cin;

  Logical_Operator3_out1 <= a OR cin;

  Logical_Operator2_out1 <= Logical_Operator3_out1 AND cin;

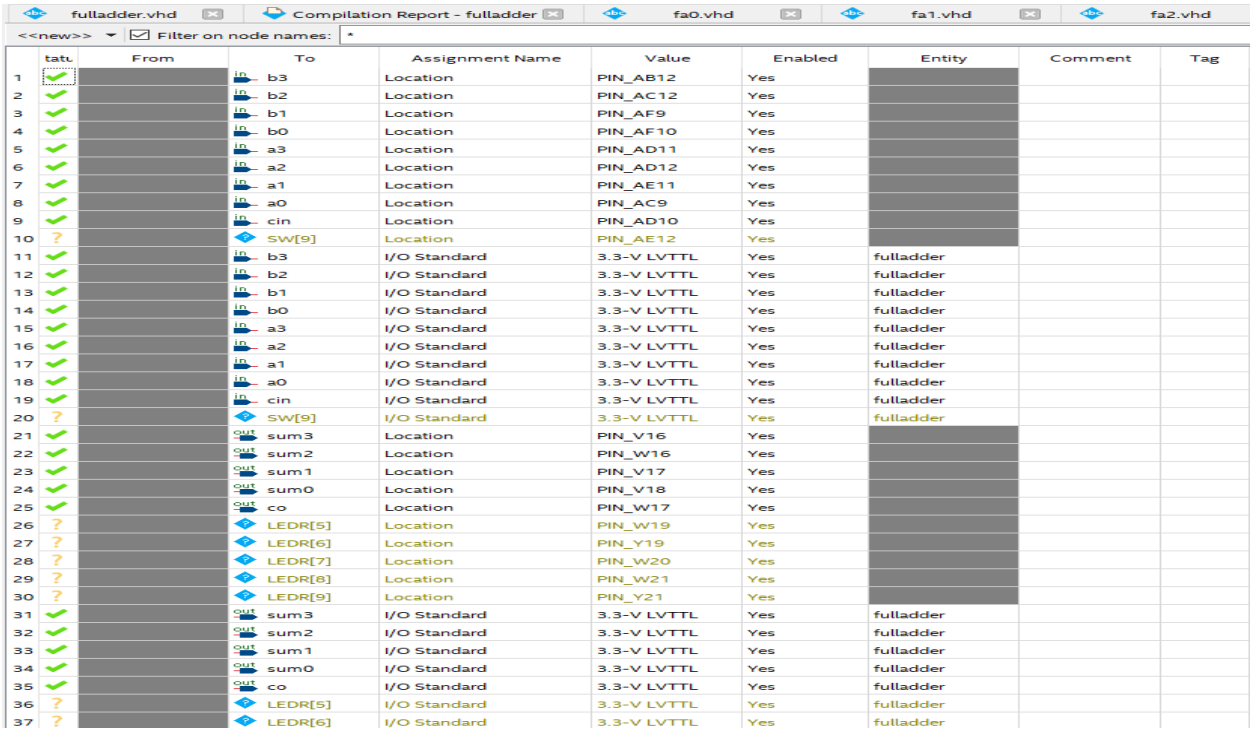
  Logical_Operator4_out1 <= Logical_Operator5_out1 OR Logical_Operator2_out1;

  sum <= Logical_Operator1_out1;

  co <= Logical_Operator4_out1;

END rtl;
```

Pin assignment for this program(below screenshot):

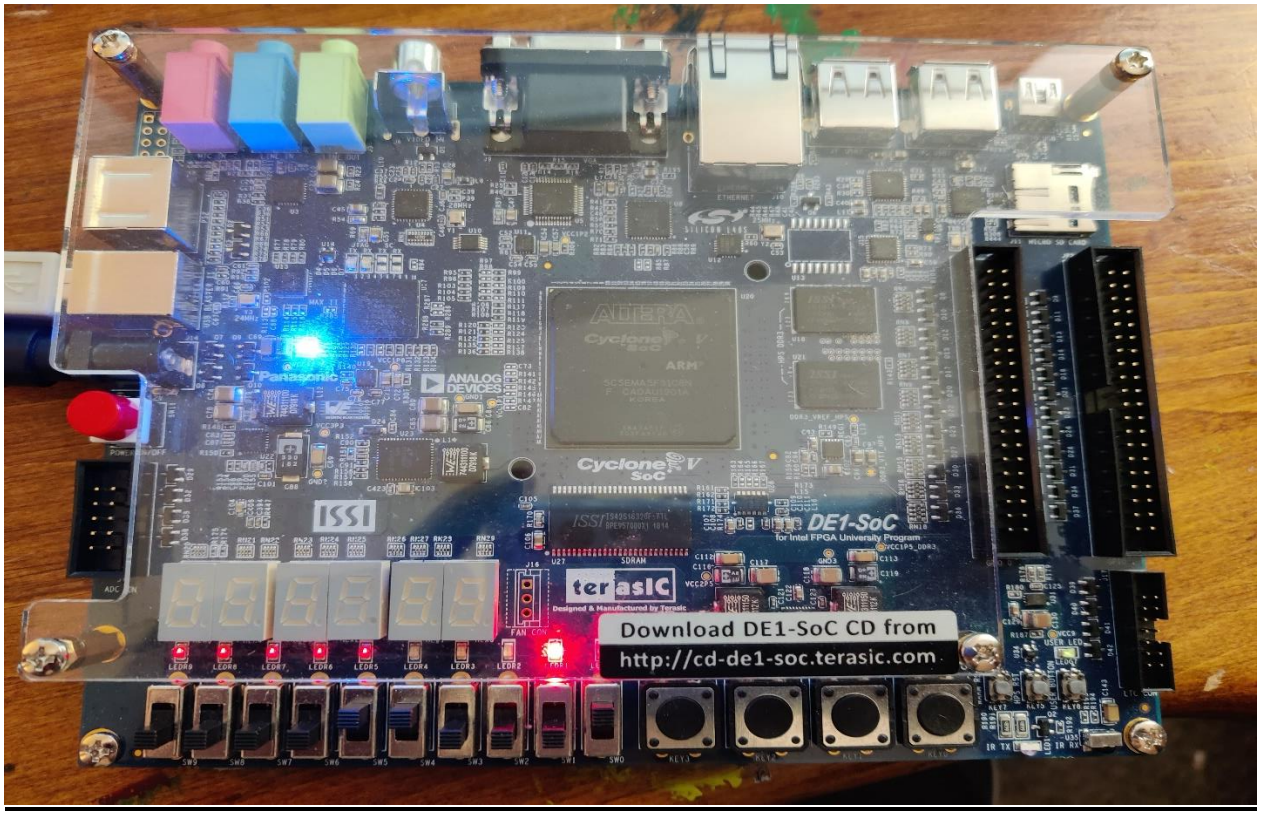


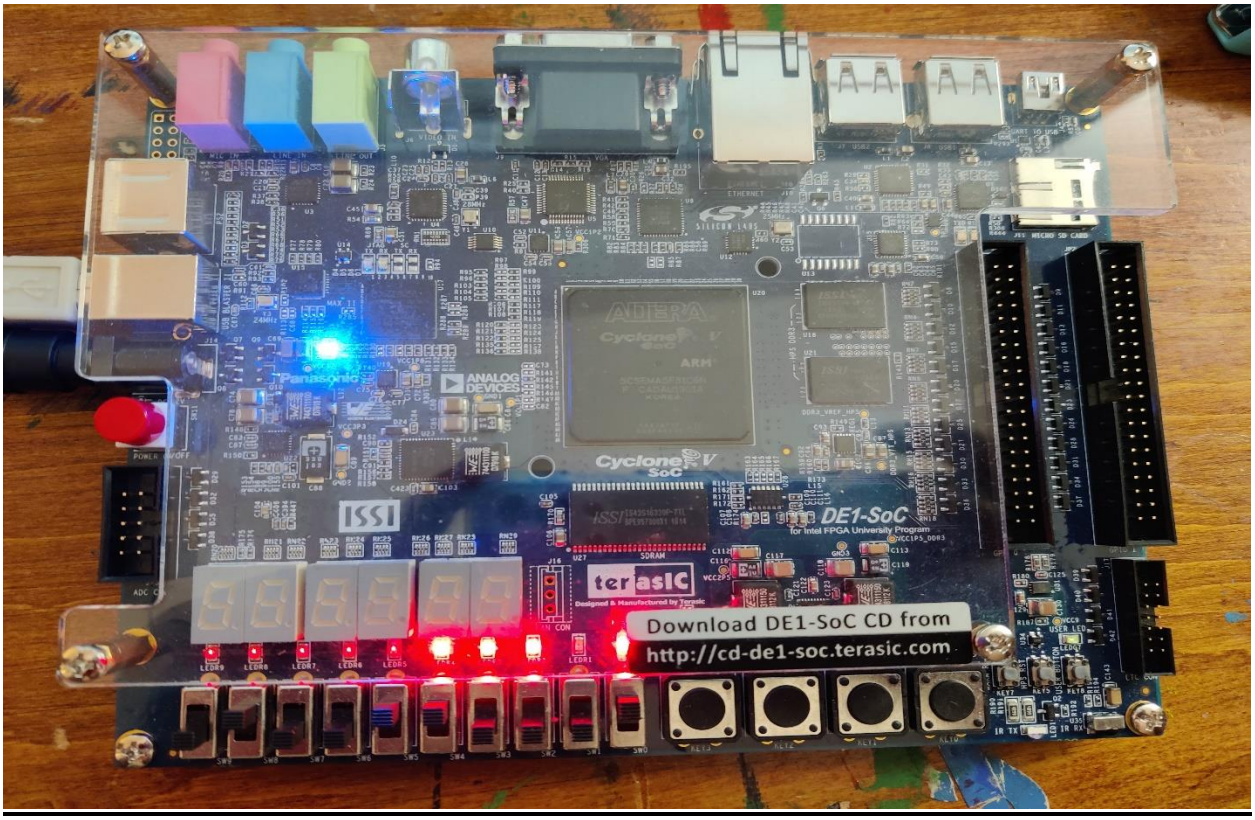
fulladder.vhd | Compilation Report - fulladder | fa0.vhd | fa1.vhd | fa2.vhd

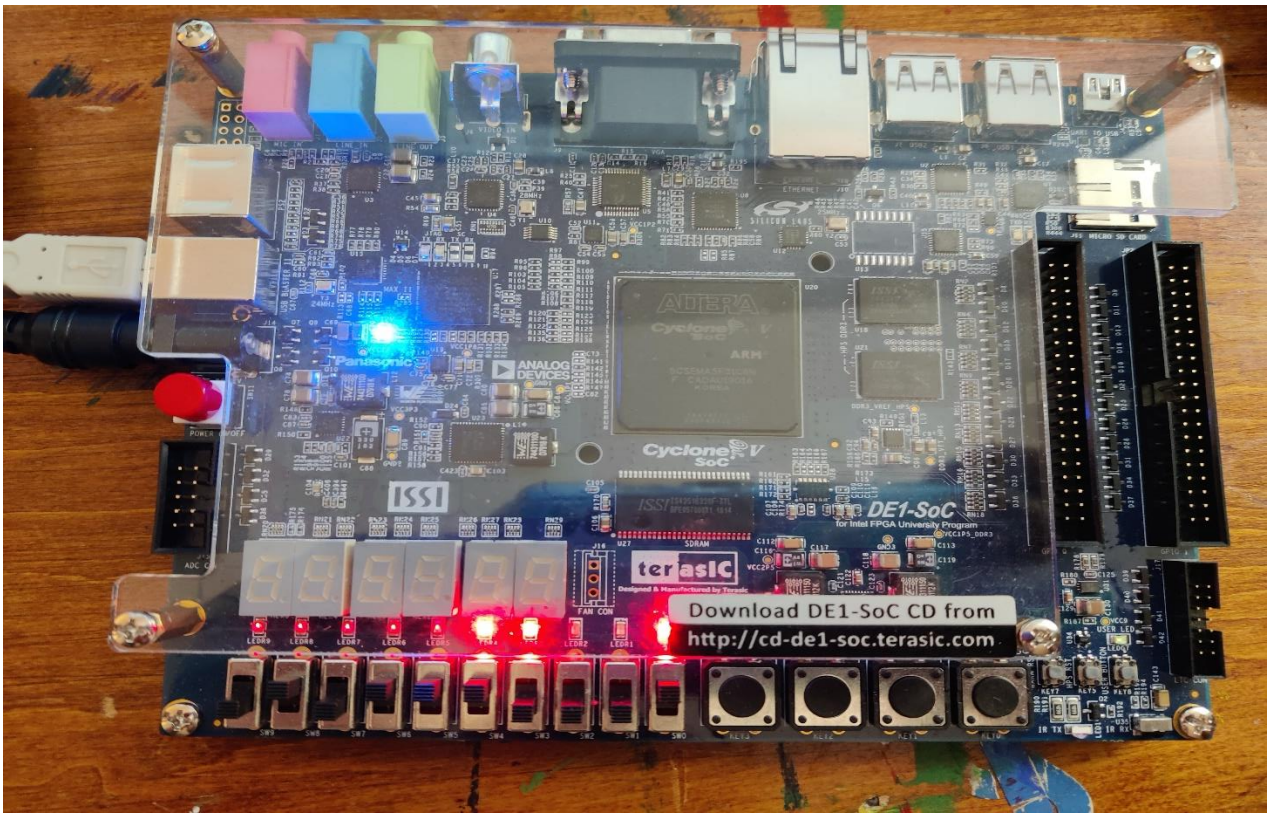
<<new>> | Filter on node names: *

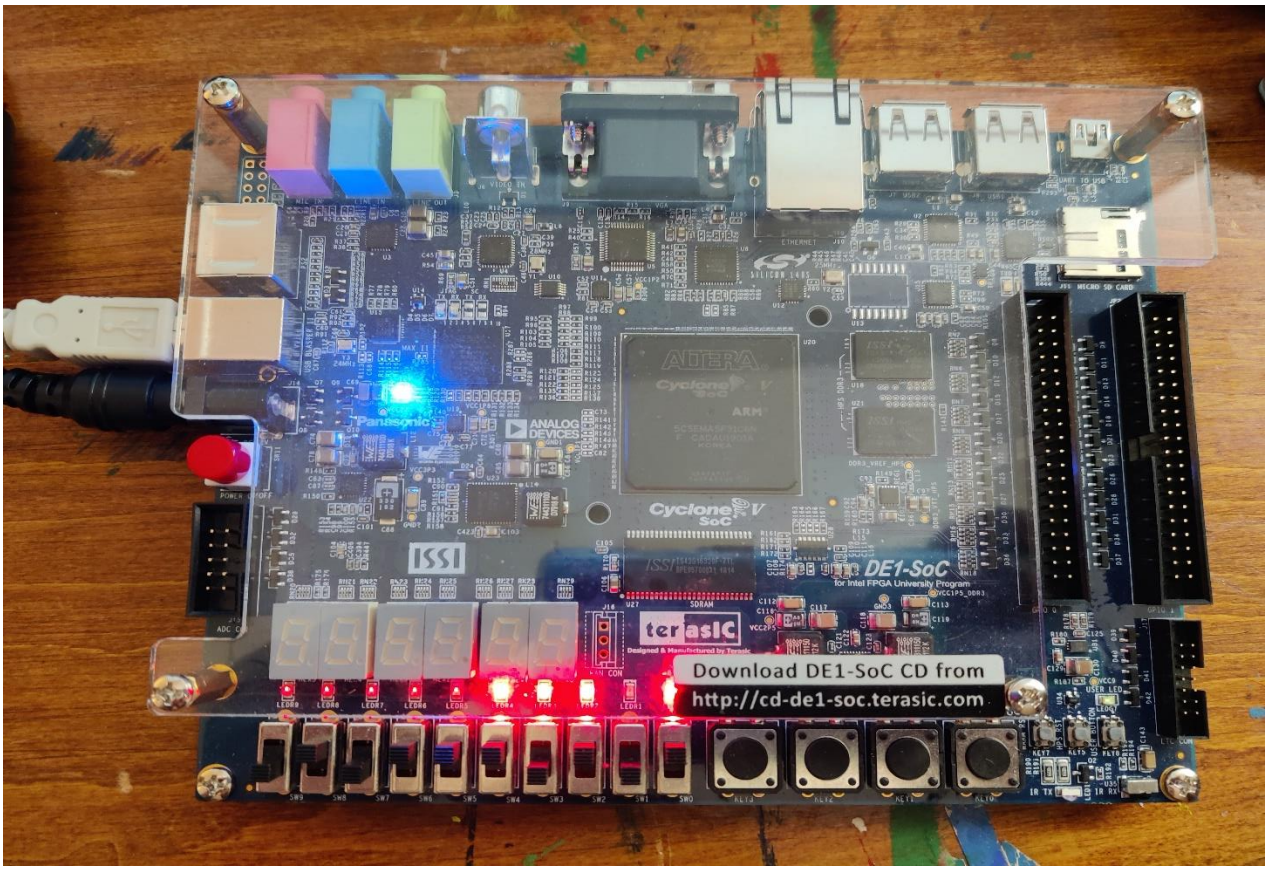
	tatu	From	To	Assignment Name	Value	Enabled	Entity	Comment	Tag
1	✓		b3	Location	PIN_AB12	Yes			
2	✓		b2	Location	PIN_AC12	Yes			
3	✓		b1	Location	PIN_AF9	Yes			
4	✓		b0	Location	PIN_AF10	Yes			
5	✓		a3	Location	PIN_AD11	Yes			
6	✓		a2	Location	PIN_AD12	Yes			
7	✓		a1	Location	PIN_AE11	Yes			
8	✓		a0	Location	PIN_AC9	Yes			
9	✓		cin	Location	PIN_AD10	Yes			
10	?		SW[9]	Location	PIN_AE12	Yes			
11	✓		b3	I/O Standard	3.3-V LVTTL	Yes	fulladder		
12	✓		b2	I/O Standard	3.3-V LVTTL	Yes	fulladder		
13	✓		b1	I/O Standard	3.3-V LVTTL	Yes	fulladder		
14	✓		b0	I/O Standard	3.3-V LVTTL	Yes	fulladder		
15	✓		a3	I/O Standard	3.3-V LVTTL	Yes	fulladder		
16	✓		a2	I/O Standard	3.3-V LVTTL	Yes	fulladder		
17	✓		a1	I/O Standard	3.3-V LVTTL	Yes	fulladder		
18	✓		a0	I/O Standard	3.3-V LVTTL	Yes	fulladder		
19	✓		cin	I/O Standard	3.3-V LVTTL	Yes	fulladder		
20	?		SW[9]	I/O Standard	3.3-V LVTTL	Yes	fulladder		
21	✓		sum3	Location	PIN_V16	Yes			
22	✓		sum2	Location	PIN_W16	Yes			
23	✓		sum1	Location	PIN_V17	Yes			
24	✓		sum0	Location	PIN_V18	Yes			
25	✓		co	Location	PIN_W17	Yes			
26	?		LEDR[5]	Location	PIN_W19	Yes			
27	?		LEDR[6]	Location	PIN_Y19	Yes			
28	?		LEDR[7]	Location	PIN_W20	Yes			
29	?		LEDR[8]	Location	PIN_W21	Yes			
30	?		LEDR[9]	Location	PIN_Y21	Yes			
31	✓		sum3	I/O Standard	3.3-V LVTTL	Yes	fulladder		
32	✓		sum2	I/O Standard	3.3-V LVTTL	Yes	fulladder		
33	✓		sum1	I/O Standard	3.3-V LVTTL	Yes	fulladder		
34	✓		sum0	I/O Standard	3.3-V LVTTL	Yes	fulladder		
35	✓		co	I/O Standard	3.3-V LVTTL	Yes	fulladder		
36	?		LEDR[5]	I/O Standard	3.3-V LVTTL	Yes	fulladder		
37	?		LEDR[6]	I/O Standard	3.3-V LVTTL	Yes	fulladder		

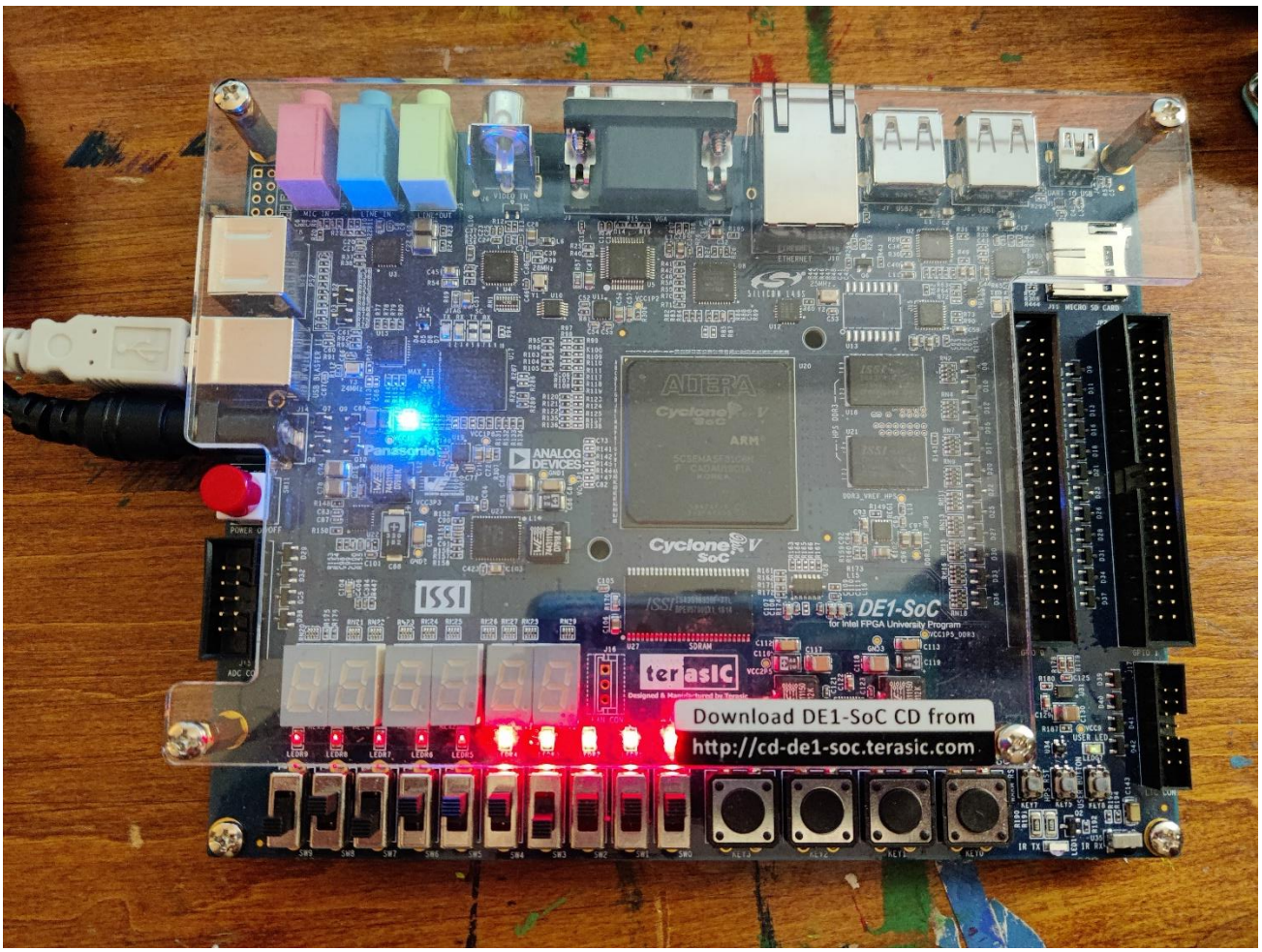
Output Screenshot: **Sample screenshots are below**






Download DE1-SoC CD from
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DE1-SoC
for Intel FPGA University Program
terasIC


Download DE1-SoC CD from
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DE1-SoC
for Intel FPGA University Program
terasIC



Note: I attach the MATLAB Simulink file inside the hw4 and hw4_fpga is the VHDL code install in the de-soc1 board.